

Revised 2026-09-10 | Deliverable No. 10 | English + 简体中文

Backend Operating Manual - ellis-visa.com

Topology

Hetzner CX23 (Falkenstein). Caddy terminates TLS for ellis-visa.com and serves the static app from /opt/ellis/src/renderer/dist; /api/* is proxied to the FastAPI backend (systemd unit ellis-backend) on 127.0.0.1:8000. Data: SQLite at /var/lib/ellis/ellis.db.

Routine operations

- Status: `systemctl status ellis-backend caddy`
- Logs: `journalctl -u ellis-backend -f (backend), -u caddy (edge)`
- Deploy a tested release pinned to its commit and artifact checksum. Verify the current database backup and retain the previous release before updating. Record the release identifier, migration steps and rollback procedure; smoke test the backend, public application and quality console after restarting.
- Health: `GET https://ellis-visa.com/api/health`

Data protection

Daily full database backup is required. `backend/scripts/backup_database.py` uses SQLite's online backup mechanism to include committed WAL content, validates the destination database and atomically publishes a private (mode 0600) file only after a successful integrity check. The daily wrapper retains the existing `/var/backups/ellis/ellis-<weekday>.db` paths. Never fall back to copying a live database file when SQLite backup fails. Record successful backups and failures; verify the deployed daily wrapper and its actual output, rather than inferring success from a schedule's existence. Keep a verified off-host copy for host-loss recovery. This is a database backup; release artifacts, configuration and necessary external storage require their own recovery arrangements.

For restore, stop the service, preserve the current state, restore the verified backup using the release-specific runbook, and check database integrity and public/quality behavior before reopening service. The contractual recovery target is at most one hour; a dated timed full recovery exercise is required to prove it. `systemd` process restart is useful but is not a disaster-recovery test. The 2026-09-10 live database-only drill restored a 23,834,624-byte backup, 99 tables and 949 route-table rows with integrity ok; database backup and restore took 0.708 seconds. The daily wrapper was updated. Full-service recovery, including host/app/configuration/secrets/DNS and off-host durability, is not certified by that database check. Details are in `AVAILABILITY.md`.

Quality-control backend

<https://ellis-visa.com/#ops> - records with combined spot-check filters (names, codes and Chinese accepted), per-record 25-field checklists, confidence and source-substantiation chips, one-click error flagging into the tracked correction queue, the change log, freshness, and the two-sheet Excel export.

The current confidence display is binary High/Low, with AI/human authorship kept in source provenance and publication controlled separately by the existing evidence/conflict gates. See

`DATA_CALIBER_MANUAL.md` for precise definitions. `info_validity` is the published policy end date only; unknown dates remain blank. `freshness_valid_until` is separate internal review-deadline metadata.

The user-approved completeness target includes documented Not applicable/Not published across all 25 fields; unknown/unsupported/pending states stay incomplete. The recorded-state metric is 81.9521% on the 2026-09-10 02:29 UTC snapshot; operational applicable 20 completeness is separately 87.32%. Retain literal 25 non-null diagnostics (72.9718% cells) as well. These are completion measurements, not certification of policy accuracy.

The current bilingual HTML/PDF manuals have been regenerated for these semantics. Old PASS labels, sample availability percentages, or unmeasured recovery claims are not current acceptance evidence.

Security posture

UFW allows 22/80/443 only. Backend binds loopback; only Caddy is exposed. Secrets ship encrypted (backend/secrets.enc, AES-256-CBC) and are decrypted on the machine at deploy time; backend/.env never enters git.

Route and product publication

QC keeps every product available for inspection. Product held and route_held are distinct: a partial route can publish a supported exemption or required e-visa while separate optional products remain withheld. The public page labels this state and excludes unsupported alternative claims from its answer and chat. Fees, filing deadlines and required conditions cannot be discarded to make a route publishable. Report default availability, fully released routes and held products separately; partial publication is not 100% product coverage.

Source: `BACKEND_OPERATING_MANUAL.md` | SHA-256
c11231763d2707034de77453de65d932f69a93ef0d99c16f2f6cbab597e7ab5d

Ellis 信息质量管控后台操作手册

本手册记录当前实现及操作要求，不代表全部数据已经核实或项目验收已经通过。

系统结构

生产架构使用 Hetzner CX23 (Falkenstein)，由 Caddy 为 ellis-visa.com 终止 TLS，并提供 /opt/ellis/src/renderer/dist 中的静态应用。/api/* 请求转发到本机 127.0.0.1:8000 的 FastAPI 服务 (systemd 服务名 ellis-backend)。SQLite 数据库路径为 /var/lib/ellis/ellis.db。

日常运行与发布

- 状态检查：systemctl status ellis-backend caddy。
- 后端日志：journalctl -u ellis-backend -f；边缘代理日志：journalctl -u caddy。
- 发布应使用经过测试、按提交与产物校验和固定的版本。更新前核实当前数据库备份，保留上一版本，并记录版本号、迁移步骤和回滚流程。重启后检查后台接口、公开应用与质控台。
- 健康检查地址：https://ellis-visa.com/api/health。

数据保护与恢复

每日必须进行完整数据库备份。backend/scripts/backup_database.py 使用 SQLite 在线备份接口，包含已提交的 WAL 内容；对备份执行完整性校验后，才以原子方式发布权限为 0600 的私有文件。每日执行脚本沿用 /var/backups/ellis/ellis-<weekday>.db 路径。

SQLite 备份失败时，不得退回到复制运行中的数据库文件。应记录备份成功与失败，核查线上每日任务及实际产物；任务存在不代表备份成功。应保存已验证的异机副本，以支持主机丢失后的恢复。此机制只备份数据库；发布产物、配置与必要的外部存储须分别安排恢复。

恢复时先停止服务并保留当前状态，按该版本的恢复流程还原已验证的备份，再检查数据库完整性、公开页面及质控功能，然后恢复服务。合同恢复目标为不超过 1 小时，是否达标须由带日期和计时记录的完整恢复演练证明。systemd 自动重启进程有帮助，但不是灾难恢复验收证明。2026-09-10 实际数据库备份与还原用时 0.708 秒，23,834,624 字节、99 张表、949 条线路表记录，完整性正常；每日任务已更新。该演练不包含主机、应用、配置、密钥、DNS 或异机副本可靠性，不能证明完整服务恢复达标。

质控台与数据解读

浏览器打开 <https://ellis-visa.com/#ops>。记录页面提供可组合的抽查筛选（支持名称、代码与中文）、逐条 25 字段清单、置信度与信源支持标识、问题标记、可追踪纠错队列、变更记录、时效信息及双工作表 Excel 导出。

质控保留全部产品供检查。产品 held 与线路 route_held 分别计数；部分发布的线路可以提供有证据的默认免签或需预先申请的电子签答案，同时拦截独立且未核实的可选签证。对客页面明确提示，并排除未核实的替代产品主张；不得为了发布而丢失费用、申请期限或入境条件。部分发布不等于 100% 产品覆盖。

当前置信度显示为 High（高）与 Low（低）两级。人工或 AI 审核身份保留在来源证据中；是否可以对客展示，由原有证据和冲突控制单独决定。详细定义见本次更新的数据口径手册。

info_validity 仅表示政策已公布的截止日期；未知日期留空。freshness_valid_until 单独表示内部复核期限。用户已同意：全部 25 字段中，有记录的不适用与未公布状态计作完整；未知、无证据或待审核仍不完整。2026-09-10 02:29 UTC 快照的记录状态完整度为 81.9521%，日常可填写 20 字段为 87.32%；全部 25 字段非空诊断为 72.9718%。这些指标不证明政策准确率。

本次版本替换旧版双语 HTML/PDF 手册。旧版 PASS 标记、示例可用率、或未计时的恢复描述，不得用作当前验收证据。

安全边界

UFW 仅开放 22/80/443 端口；后端仅绑定本机回环地址，由 Caddy 对外提供服务。部署资料中的密钥以 backend/secrets.enc 加密保存（AES-256-CBC），并在部署主机解密；backend/.env 不进入 Git。手册不包含凭据。

本文依据当前实现编制；最终验收仍需实际证据。